```
#include <stdio.h>
int gcd(int a, int b) {
    while (b) {
        int t = b;
        b = a % b;
        a = t;
    }
    return a;
}
int lcm(int a, int b) {
    return (a * b) / gcd(a, b);
}
struct Task {
    int id;
    int burst;
    int period;
    int remaining;
};

int main() {
    int n;
    printf("1WA24CS217\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    struct Task t[n];
    printf("Enter burst times:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &t[i].burst);
        t[i].id = i + 1;
    }
```

```
printf("Enter periods:\n");
for (int i = 0; i < n; i++) {
    scanf("%d", &t[i].period);
    t[i].remaining = 0;
}

for (int i = 0; i < n - 1; i++) {
    for (int j = i + 1; j < n; j++) {
        if (t[i].period > t[j].period) {
            struct Task temp = t[i];
            t[i] = t[j];
            t[j] = temp;
        }
    }
}

int hyper = t[0].period;
for (int i = 1; i < n; i++)
    hyper = lcm(hyper, t[i].period);
float util = 0;
for (int i = 0; i < n; i++)
    util += (float)t[i].burst / t[i].period;

float bound;
if (n == 1) bound = 1.0;
else if (n == 2) bound = 0.828427;
else if (n == 3) bound = 0.779763;
else bound = 0.693;
printf("\n%f <= %f => %s\n", util, bound, (util <= bound) ? "true" : "false");
```

```
printf("\nScheduling occurs for %d ms\n\n", hyper);

int remaining[n];
for (int i = 0; i < n; i++)
    remaining[i] = 0;

int last = -2;

for (int time = 0; time < hyper; time++) {

    for (int i = 0; i < n; i++) {
        if (time % t[i].period == 0)
            remaining[i] = t[i].burst;
    }

    int run = -1;
    for (int i = 0; i < n; i++) {
        if (remaining[i] > 0) {
            run = i;
            break;
        }
    }

    if (run != last) {
        if (run != -1)
            printf("%dms onwards: Process %d running\n", time, t[run].id);
        else
            printf("%dms onwards: CPU is idle\n", time);
    }
```

```
        if (run != -1)
            remaining[run]--;


        last = run;
    }


    return 0;
}
```

OUTPUT:

```
1WA24CS217
Enter number of processes: 3
Enter burst times:
3 2 2
Enter periods:
20 5 10

0.750000 <= 0.779763 => true

Scheduling occurs for 20 ms

0ms onwards: Process 2 running
2ms onwards: Process 3 running
4ms onwards: Process 1 running
5ms onwards: Process 2 running
7ms onwards: Process 1 running
9ms onwards: CPU is idle
10ms onwards: Process 2 running
12ms onwards: Process 3 running
14ms onwards: CPU is idle
15ms onwards: Process 2 running
17ms onwards: CPU is idle
```